You can remove `np.int64` **by converting them to native ints** before printing.

✅ Best / simplest fix

```
labels = tuple(int(x) for x in labels)
print("labels:", labels)
```

Output:

```
labels: (2, 1)
```

Why this happens

NumPy arrays and operations often return **NumPy scalar types**:

```
type(np.int64(2))  # <class 'numpy.int64'>
```

When printed inside containers (tuple, list), NumPy keeps the type name.

Other clean options (depending on context)

1. Convert via `.item()`

```
labels = tuple(x.item() for x in labels)
```

---

2. Convert using `.astype(int)` (if it's an array)

```
labels = tuple(labels.astype(int))
```

---

3. If you only care about printing (not data)

```
print("labels:", tuple(map(int, labels)))
```

---

4. If `labels` is a NumPy array

```
labels = labels.tolist()
print("labels:", tuple(labels))
```

❌ What NOT to do

- Don’t rely on string replacement like:
- `str(labels).replace("np.int64", "")`

  (brittle and unsafe)

Rule of thumb

Convert NumPy scalars to Python scalars at API boundaries
(printing, logging, JSON, UI, filenames).